Experiment 2

Code:

```
# Candidate Elimination Algorithm
data = [
    ['Sunny','Warm','Normal','Strong','Warm','Same','Yes'],
    ['Sunny','Warm','High','Strong','Warm','Same','Yes'],
    ['Rainy','Cold','High','Strong','Warm','Change','No'],
    ['Sunny','Warm','High','Strong','Cool','Change','Yes']
]
S = data[0][:-1]
G = [['?' for _ in range(len(S))]]
for row in data:
    x = row[:-1]
    y = row[-1]
    if y == 'Yes':
        for i in range(len(S)):
            if S[i] != x[i]:
                S[i] = '?'
    else:
        new_G = []
        for g in G:
            for i in range(len(S)):
                if g[i] == '?' and S[i] != x[i]:
                    h = g.copy()
                    h[i] = S[i]
                    new_G.append(h)
        G = new_G
print("Final Specific Hypothesis (S):", S)
print("Final General Hypothesis (G):")
for g in G:
```

print(g)

Output:

```
    else:
        new_G = []
        for g in G:
            for i in range(len(S)):
                if g[i] == '?' and S[i] != x[i]:
                    h = g.copy()
                    h[i] = S[i]
                    new_G.append(h)
        G = new_G

print("Final Specific Hypothesis (S):", S)
print("Final General Hypothesis (G):")
for g in G:
    print(g)
```

```
Final Specific Hypothesis (S): ['Sunny', 'Warm', '?', 'Strong', '?', '?']
Final General Hypothesis (G):
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', 'Same']
```